LECTURE-25 EXPONENTIAL FOURIER SERIES

Mutually Orthogonal Complex Exponential

- Harmonically related complex exponential functions are orthogonal to each other.

$$S = \left\{ e^{jk\omega_0 t} \right\}_{k=-\infty}^{\infty}$$

$$\int_0^{2\pi/\omega_0} e^{jm\omega_0 t} \left(e^{jn\omega_0 t} \right)^* dt$$

$$= \int_0^{2\pi/\omega_0} e^{j(m-n)\omega_0 t} dt$$

$$= \left[\frac{e^{j(m-n)\omega_0 t}}{j(m-n)\omega_0} \right]_0^{2\pi/\omega_0}$$

$$= \frac{e^{j2\pi(m-n)} - 1}{j(m-n)\omega_0} = \begin{cases} 0, & m \neq n \\ 2\pi/\omega_0, & m = n \end{cases}$$

$$= \begin{cases} 0, & m \neq n \\ T_0, & m = n \end{cases}$$

Complex exponential of diff. f. that are harmonically related are orthogonal.

Complex exponential Fourier Series

- A non-sinusoidal periodic signal can be expressed as linear combination of harmonically related complex exponentials.

$$\boxed{\begin{array}{l} x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \\ C_n : \text{exponential FS coefficient} \end{array}}$$

** SN

Computing C_K

$$x(t)\, e^{-jK\omega_0 t} = e^{-jK\omega_0 t} \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$= \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \cdot e^{-jK\omega_0 t}$$

$$\int_0^T x(t) e^{-jK\omega_0 t} dt = \sum_{n=-\infty}^{\infty} C_n \underbrace{\int_0^T e^{jn\omega_0 t} e^{-jK\omega_0 t} dt}_{\text{Inner product of } e^{jn\omega_0 t},\ e^{jK\omega_0 t}}$$

$$\int_0^T x(t) e^{-jK\omega_0 t} dt = C_K \cdot T$$

$$C_K = \frac{1}{T} \int_0^T x(t) e^{-jK\omega_0 t} dt$$

$$\boxed{C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt}$$

** SN

$$C_0 = a_0 = \frac{1}{T} \int_0^T x(t) dt$$

Magnitude & Phase Spectra of periodic signals

- A plot of $|C_n|$ Vs n ; ie magnitude of C_n vs n or nf_0 or f is called Magnitude Spectra.
- A plot of $\angle C_n$ vs n i.e. phase of C_n vs n or nf_0 or f is called Phase Spectra.
- For a real signal, magnitude spectrum has even symmetry while phase spectrum has odd symmetry.

**SN

$$C_n = \frac{1}{T}\int_{\langle T\rangle} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$C_n^* = \frac{1}{T}\int_{\langle T\rangle} x^*(t)\, e^{jn\omega_0 t}\, dt$$

$x(t)$: real $\quad x^*(t) = x(t)$

$$C_n^* = \frac{1}{T}\int_{\langle T\rangle} x(t)\, e^{jn\omega_0 t}\, dt = C_{-n}$$

$$C_{-n} = C_n^*$$

$|C_{-n}| = |C_n^*| = |C_n|$: even magnitude

$\angle C_{-n} = -\angle C_n$: odd phase

Important features

- Generally, FS expansion of $x(t)$ involves an infinite number of coefficients. However, if $x(t)$ is band limited signal the number of FS coefficient are finite.

- In case of non band-limited signal we try to approximate $x(t)$ by $x_N(t)$

$$x_N(t) \triangleq \sum_{n=-N}^{N} C_n e^{jn\omega_0 t}, \quad N: \text{integer}$$

as $N \to \infty$ approximation gets better.

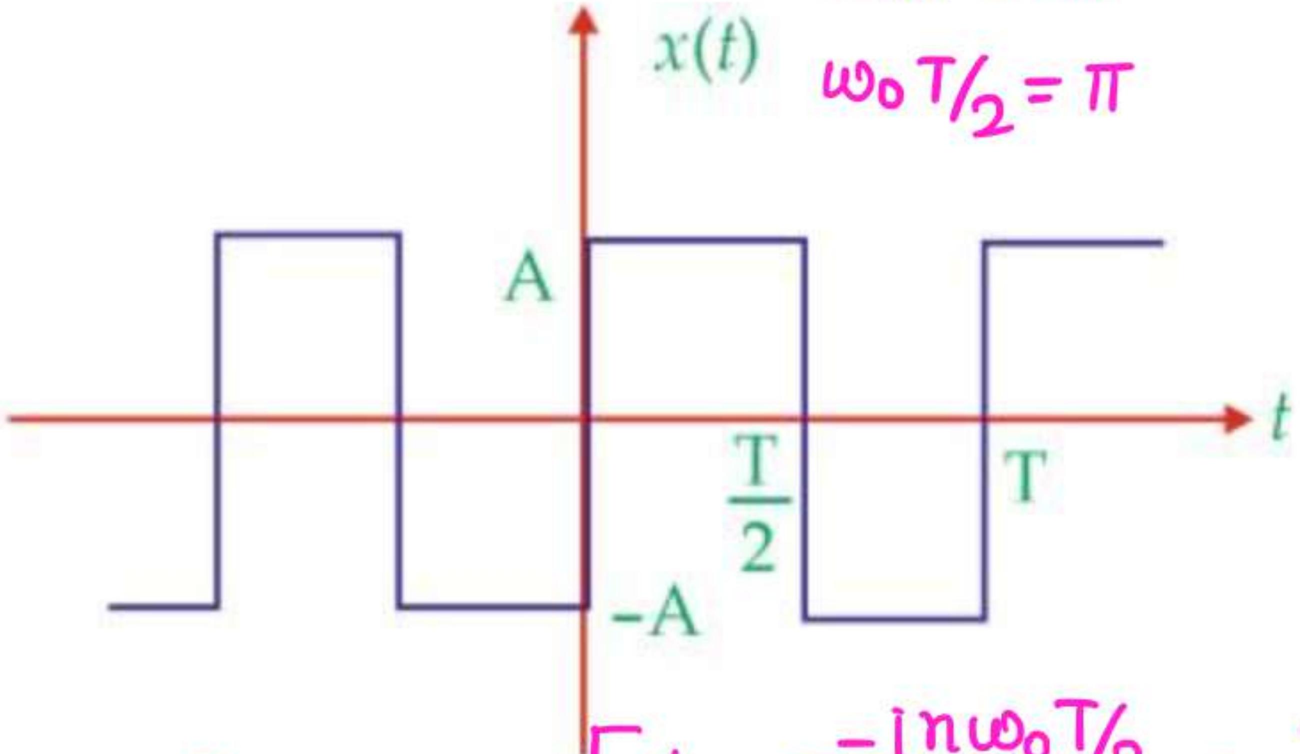
- projection of $x(t)$ a vector in infinite dimensional space on a N-dimensional subspace.

Question-01

Find complex exponential Fourier Series expansion & plot magnitude & phase spectra for a square wave.

$\omega_0 T = 2\pi$

$\omega_0 T/2 = \pi$



$$c_n = \frac{1}{T}\int_0^T x(t)\, e^{-jn\omega_0 t}\,dt$$

$$= \frac{1}{T}\left[\int_0^{T/2} A\, e^{-jn\omega_0 t}\,dt - \int_{T/2}^{T} A\, e^{-jn\omega_0 t}\,dt\right]$$

$$= \frac{A}{T}\left[\left(\frac{e^{-jn\omega_0 t}}{-jn\omega_0}\right)_0^{T/2} + \left(\frac{e^{-jn\omega_0 t}}{jn\omega_0}\right)_{T/2}^{T}\right]$$

$$c_n = \frac{A}{T(jn\omega_0)}\left[1 - e^{-jn\omega_0 T/2} + e^{-jn\omega_0 T} - e^{-jn\omega_0 T/2}\right]$$

$$= \frac{A}{j2n\pi}\left[1 - e^{-jn\pi} + e^{-j2n\pi} - e^{-jn\pi}\right] = \frac{A}{j2n\pi}\left[2 - 2e^{-jn\pi}\right]$$

$$C_n = \frac{A}{jn\pi}\left[1 - e^{-jn\pi}\right]$$

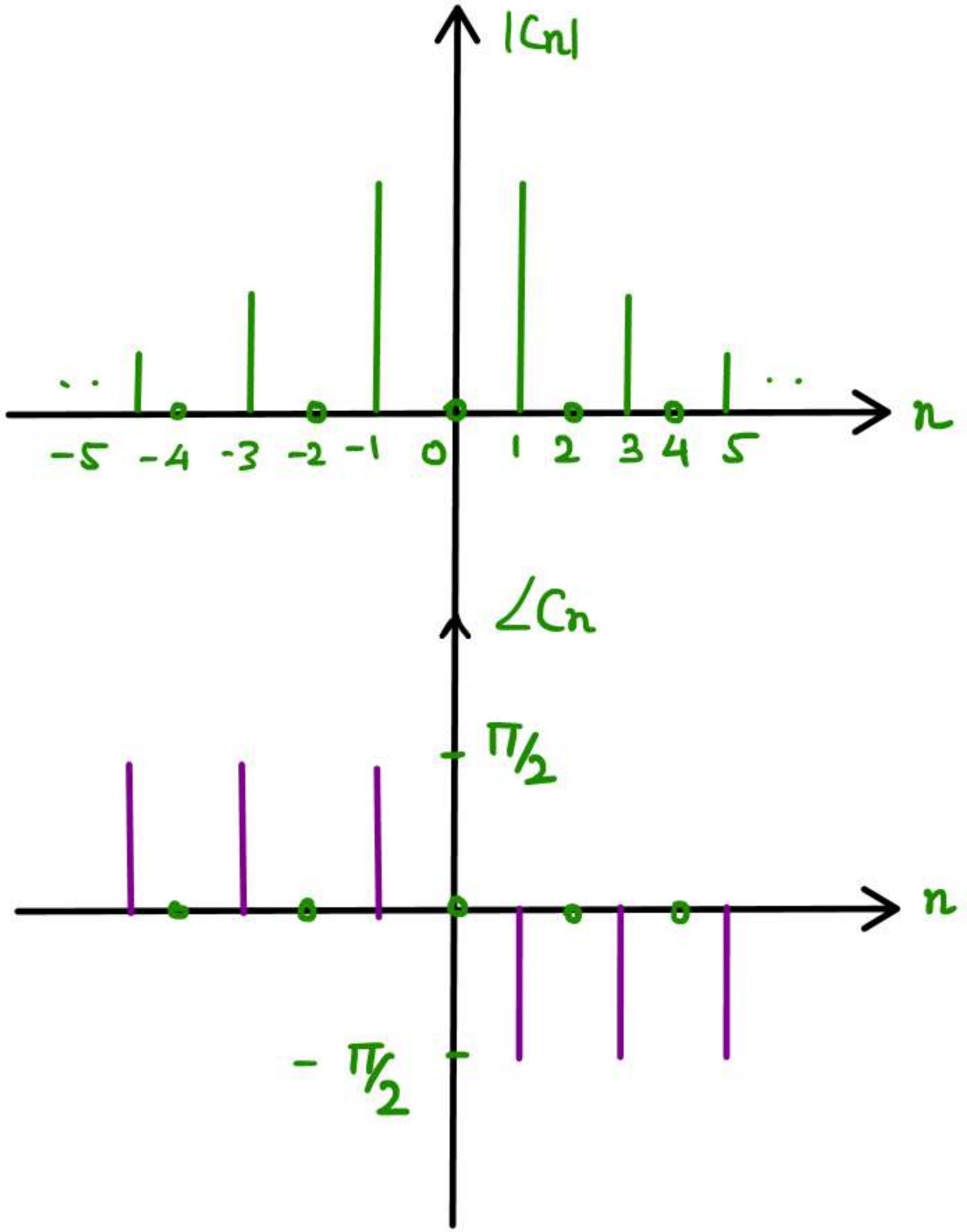
$$e^{-jn\pi} = (-1)^n = \begin{cases} 1, & n = \text{even} \\ -1, & n = \text{odd} \end{cases}$$

$$C_n = \frac{2A}{jn\pi}, \quad n = \text{odd}, \quad C_n = 0, \quad n = \text{even}$$

$$x(t) = \sum_{\substack{n=-\infty \\ n = \text{odd}}}^{\infty} \frac{2A}{jn\pi} e^{jn\omega_0 t}$$

$$|C_n| = \left|\frac{2A}{n\pi}\right|$$

$$\angle C_n = \angle \frac{2A}{jn\pi} = \begin{cases} -\pi/2, & n > 0 \\ \pi/2, & n < 0 \end{cases}$$



Relation between Trigonometric & Exponential FS Coefficients

$$c_n = \frac{1}{T}\int_{<T>} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T}\int_{<T>} x(t)\left[\cos n\omega_0 t - j\sin n\omega_0 t\right] dt$$

$$= \frac{1}{T}\int_{<T>} x(t)\cos n\omega_0 t\, dt - j\frac{1}{T}\int_{<T>} x(t)\sin n\omega_0 t\, dt$$

$$= \frac{a_n}{2} - j\frac{b_n}{2} = \frac{a_n - jb_n}{2}$$

$$c_{-n} = \frac{1}{T}\int_{<T>} x(t)\, e^{jn\omega_0 t}\, dt = \frac{1}{T}\int_{<T>} x(t)\left[\cos n\omega_0 t + j\sin n\omega_0 t\right] dt$$

$$= \frac{1}{T}\int_{<T>} x(t)\cos n\omega_0 t\, dt + j \times \frac{1}{T}\int_{<T>} x(t)\sin n\omega_0 t\, dt = \frac{a_n + jb_n}{2}$$

Relation between Trigonometric & Exponential FS Coefficients

$C_n + C_{-n} = a_n$

$C_n - C_{-n} = -j b_n$

$b_n = j(C_n - C_{-n})$

Summary

$a_n = C_n + C_{-n}$

$b_n = j(C_n - C_{-n})$

$C_n = \frac{a_n - j b_n}{2}$

** SN

$|C_n| = \frac{\sqrt{a_n^2 + b_n^2}}{2} = r_n/2$

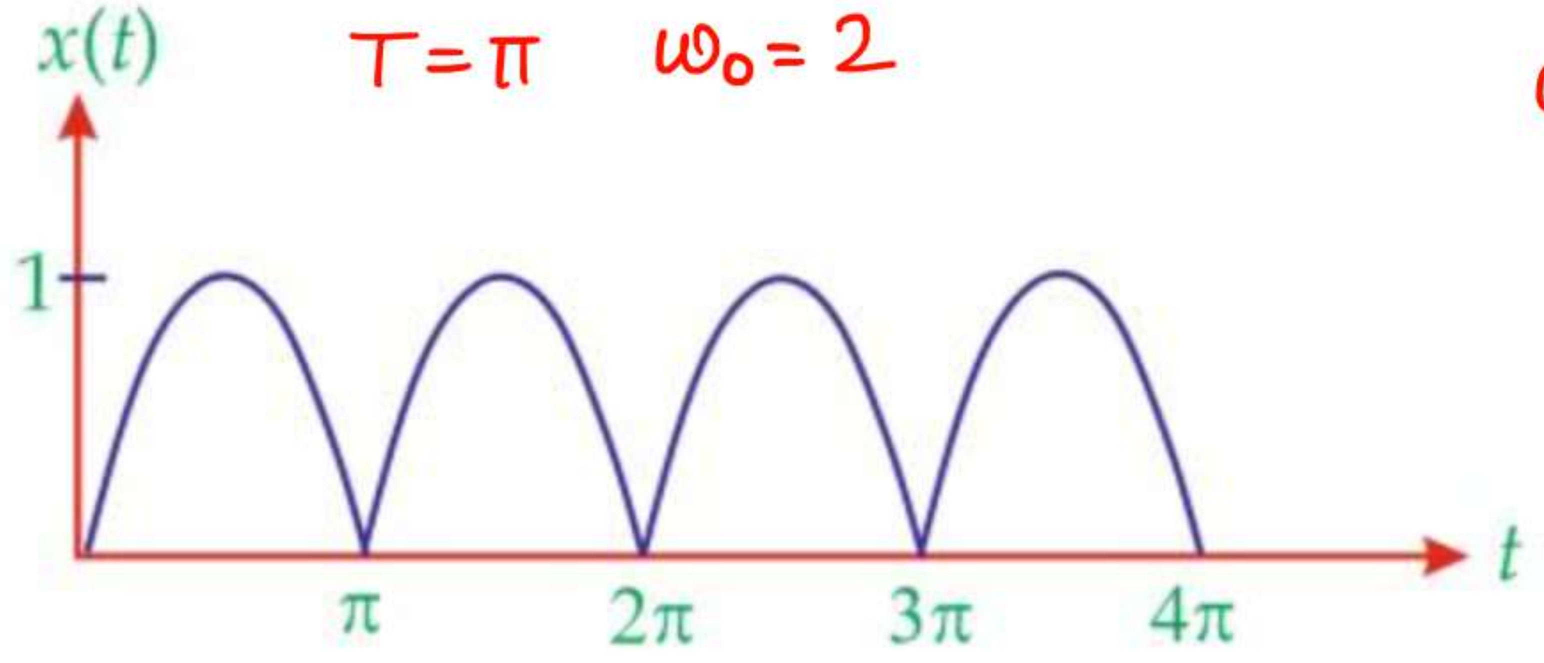
$r_n = 2|C_n|$

$\angle C_n = -\tan^{-1} b_n/a_n$

Question-02

Determine exponential Fourier Series for following signals & plot the spectrum.

(1) $x(t) = |\sin t|$



$T = \pi \quad \omega_0 = 2$

$x(t) = \sin t \qquad 0 < t < \pi$

$$C_n = \frac{1}{T}\int_0^{\pi} \sin t \cdot e^{-jn\omega_0 t}\,dt$$

$$= \frac{1}{T}\int_0^{\pi} \left(\frac{e^{jt} - e^{-jt}}{2j}\right) e^{-jn\omega_0 t}\,dt$$

$$= \frac{1}{2jT}\left[\int_0^{\pi} \left(e^{-j(2n-1)t} - e^{-j(2n+1)t}\right)dt\right]$$

$$C_n = \frac{1}{2j\pi}\left[\frac{e^{-j(2n-1)t}}{-j(2n-1)} - \frac{e^{-j(2n+1)t}}{-j(2n+1)}\right]_0^{\pi}$$

$$c_n = \frac{1}{2j\pi}\left[\frac{e^{-j(2n-1)\pi}}{-j(2n-1)} - \frac{e^{-j(2n+1)\pi}}{-j(2n+1)} - \frac{1}{-j(2n-1)} + \frac{1}{-j(2n+1)}\right]$$

$$= \frac{1}{2j\pi}\left[\frac{1}{j(2n-1)} - \frac{1}{j(2n+1)} + \frac{1}{j(2n-1)} - \frac{1}{j(2n+1)}\right]$$

$$= \frac{2}{2j^2\pi}\left[\frac{1}{(2n-1)} - \frac{1}{(2n+1)}\right]$$

$$= -\frac{1}{\pi}\left[\frac{2}{4n^2-1}\right] = \frac{2}{\pi(1-4n^2)}$$

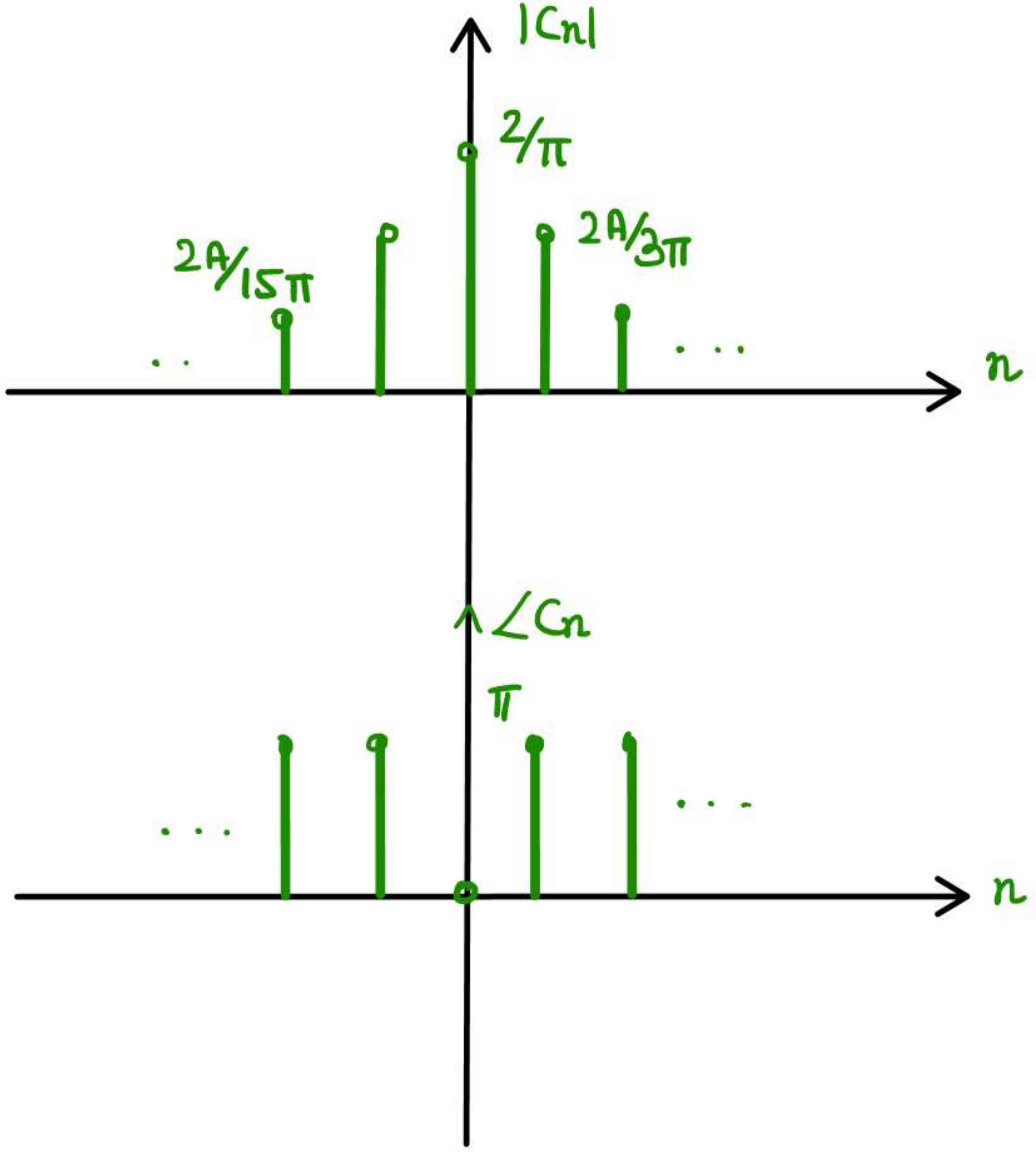
$n = 0 \quad c_0 = 2/\pi$

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{2}{\pi(1-4n^2)} e^{jn\omega_0 t}$$

$$C_n = \frac{2}{\pi(1-4n^2)}$$

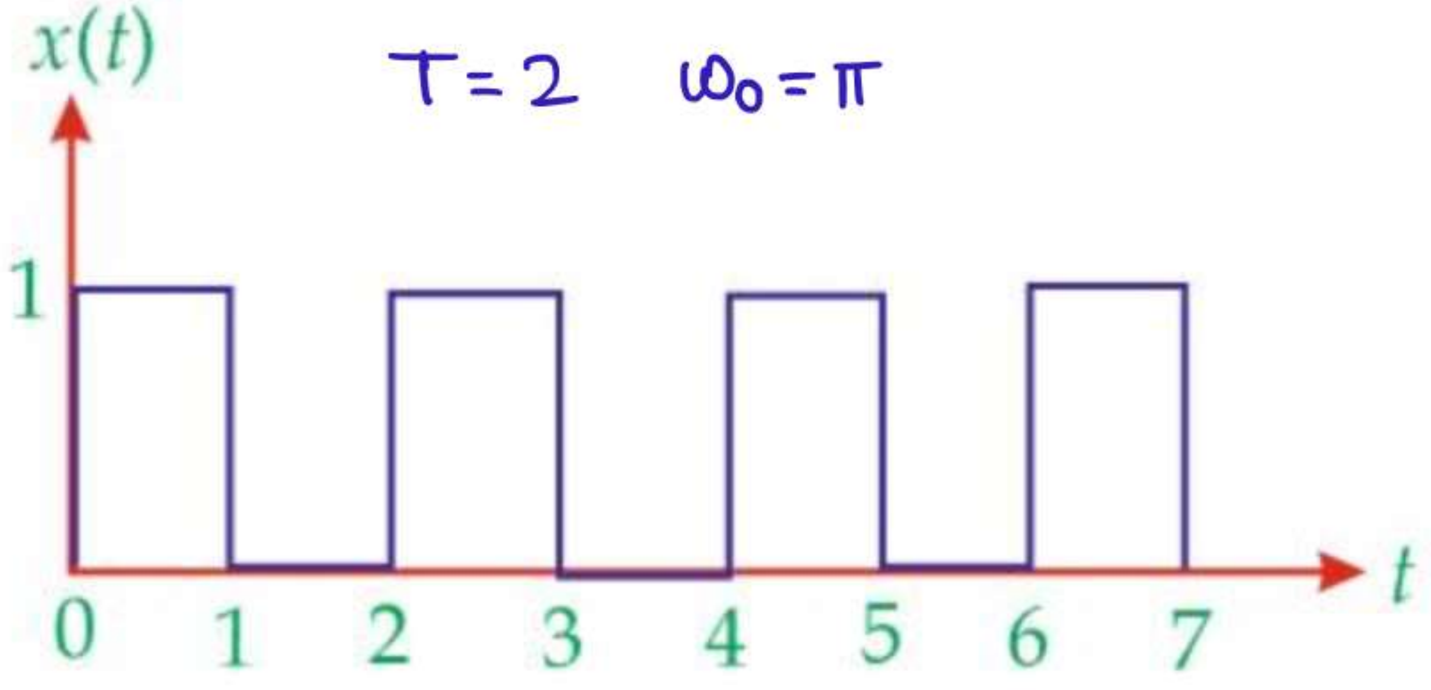
$$|C_n| = \frac{2}{\pi\,|1-4n^2|}$$

$$\angle C_0 = 0^\circ \qquad \angle C_n = \pi \quad \forall\ n \neq 0$$



Question-03

Determine exponential Fourier Series for following signals & plot the spectrum.



$T = 2 \quad \omega_0 = \pi$

$$c_n = \frac{1}{2}\int_0^1 e^{-jn\omega_0 t}\,dt$$

$$= \frac{1}{2}\left[\frac{e^{-jn\omega_0 t}}{-jn\omega_0}\right]_0^1 = \frac{1}{2jn\omega_0}\left[1 - e^{-jn\omega_0}\right]$$

$$c_n = \frac{1}{2jn\pi}\left(1 - e^{-jn\pi}\right)$$

$$e^{-jn\pi} = (-1)^n = \begin{cases} 1, & n = \text{even} \\ -1, & n = \text{odd} \end{cases}$$

$$c_n = \frac{2}{2jn\pi} = \frac{1}{jn\pi} \quad \forall\ n = \text{odd}$$

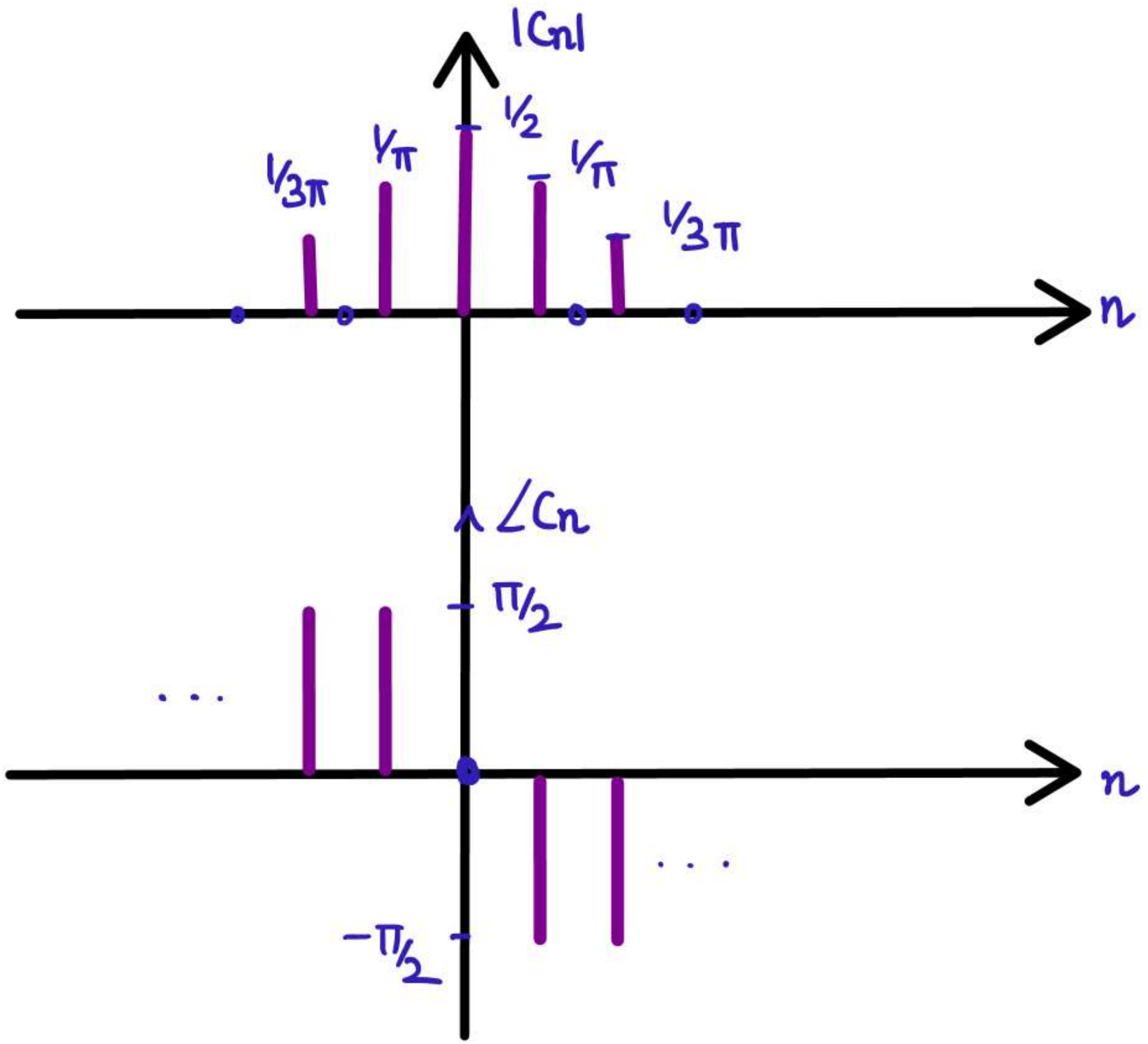
$$c_n = 0 \quad \forall\ n = \text{even}$$

$$c_0 = \frac{1}{2}$$

$$C_n = \frac{1}{jn\pi}$$

$$|C_n| = \frac{1}{|n|\pi}$$

$$\angle C_n = \begin{cases} -\pi/2 , & n>0 \\ \pi/2 , & n<0 \end{cases}$$



Question-04

Compute exponential Fourier Series Coefficients

(1) $x(t)=\cos \omega_0 t$

(2) $x(t)=\sin \omega_0 t$

$$x(t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$$

$$\Rightarrow c_1 = \frac{1}{2}$$

$$c_{-1} = \frac{1}{2}$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

$$x(t) = \frac{e^{+j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$= \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$$

$$c_1 = \frac{1}{2j} \qquad c_{-1} = \frac{-1}{2j}$$

Question-05

Compute exponential Fourier Series Coefficients

(i) $x(t)=\sin 2t + \cos 3t$

(ii) $x(t)=\sin(3t-\pi/3)$ $\quad \omega_0 = 3$

(i) $\omega_0 = \text{HCF}(2,3) = 1$

$$x(t) = \frac{e^{j2t} - e^{-j2t}}{2j} + \frac{e^{j3t} + e^{-j3t}}{2}$$

$$= \frac{1}{2j} e^{j2(1)t} - \frac{1}{2j} e^{-2j(1)t}$$

$$+ \frac{1}{2} e^{j3(1)t} + \frac{1}{2} e^{j(-3)(1)t}$$

$C_2 = 1/2j \quad C_{-2} = 1/2j \quad C_3 = 1/2 \quad C_{-3} = 1/2$

(ii) $$x(t) = \frac{e^{j(3t-\pi/3)} - e^{-j(3t-\pi/3)}}{2j}$$

$$= e^{j3t}\left(\frac{e^{-j\pi/3}}{2j}\right)$$

$$- e^{-j3t}\left(\frac{e^{j\pi/3}}{2j}\right)$$

$$C_1 = \frac{e^{-j\pi/3}}{2j}$$

$$C_{-1} = \frac{e^{j\pi/3}}{2j}$$

Question-06

$\rightarrow C_5, C_{-5}$ $\rightarrow C_6, C_{-6}$ $\rightarrow C_0$

For a signal x(t) = 5sin5t + 7cos6t + 3, the non-zero coefficients of exponential Fourier series are

$\omega_0 = HCF(5, 6) = 1$

(a) $C_{-6}, C_{-5}, C_0, C_5, C_6,$

(b) $C_{-3}, C_{-1}, C_0, C_1, C_3,$

(c) $C_{-2}, C_{-1}, C_0, C_1, C_2,$

(d) $C_{-4}, C_{-2}, C_0, C_2, C_4,$

Question-07

The Fourier series expansion of a real periodic signal with fundamental frequency f_0 is given by $g_p(t) = \sum_{n=-x}^{\infty} C_n\, exp(j2\pi n f_0 t)$. It is given that $C_3 = 3 + j5$ then C_{-3} is

(A) 5+j 3

(B) -3-j 5

(C) -5+j 3

(D) 3-j 5

real signal $C_n^* = C_{-n}$

$C_{-3} = C_3^* = 3 - j5$

Question-08

The Fourier Series representation of an impulse train denoted by $s(t) = \sum_{n=-\infty}^{\infty} \delta\,(t - nT_0)$ is given by :

(A) $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} exp^{\left(\frac{-j2\pi nt}{T_0}\right)}$

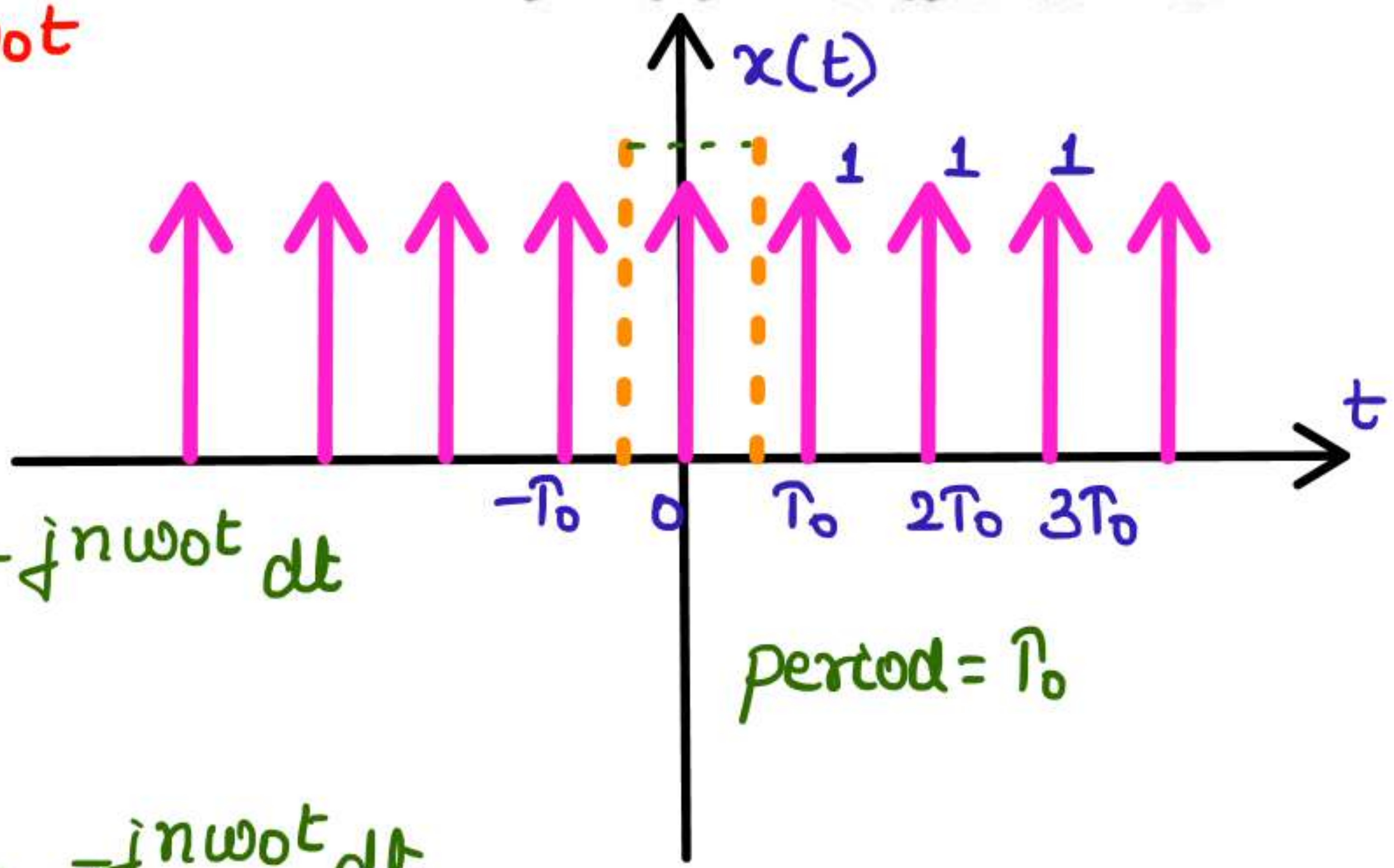
(B) $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} exp^{\left(\frac{-j\pi nt}{T_0}\right)}$

(C) $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} exp^{\left(\frac{j\pi nt}{T_0}\right)}$

(D) $\frac{1}{T_0}\sum_{n=-\infty}^{\infty} exp^{\left(\frac{j2\pi nt}{T_0}\right)}$

$$x(t) = \frac{1}{T_0}\sum_{n=-\infty}^{\infty} e^{+jn\omega_0 t}$$

$$\omega_0 = 2\pi/T_0$$



$$C_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \delta(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T_0}\left[e^{-jn\omega_0 t}\right]\Big|_{t=0} = 1/T_0$$

Question-09

** SN

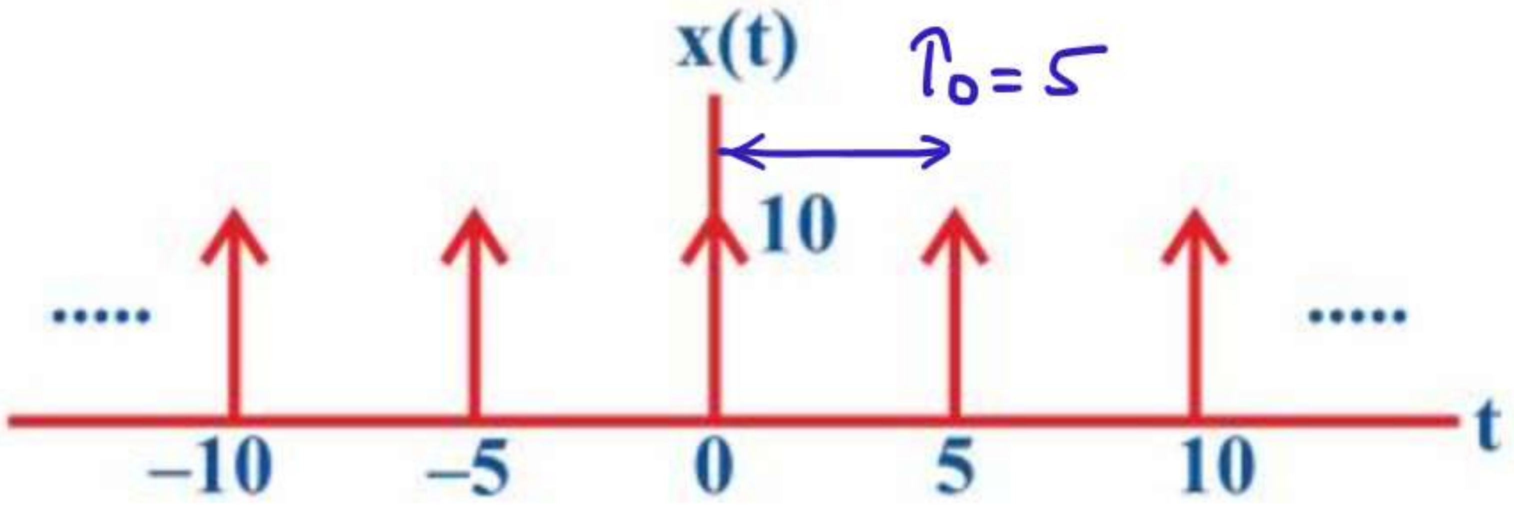
The exponential Fourier series coefficient for the periodic signal shown below is

(A) 1

(B) $cos\left(\frac{\pi}{2}n\right)$

(C) $sin\left(\frac{\pi}{2}n\right)$

~~(D)~~ 2



impulse train

all FS coeff are equal $= A/T_0 = 10/5 = 2$

A : strength of impulse

Question-10

The exponential Fourier series coefficient for the periodic signal shown below is

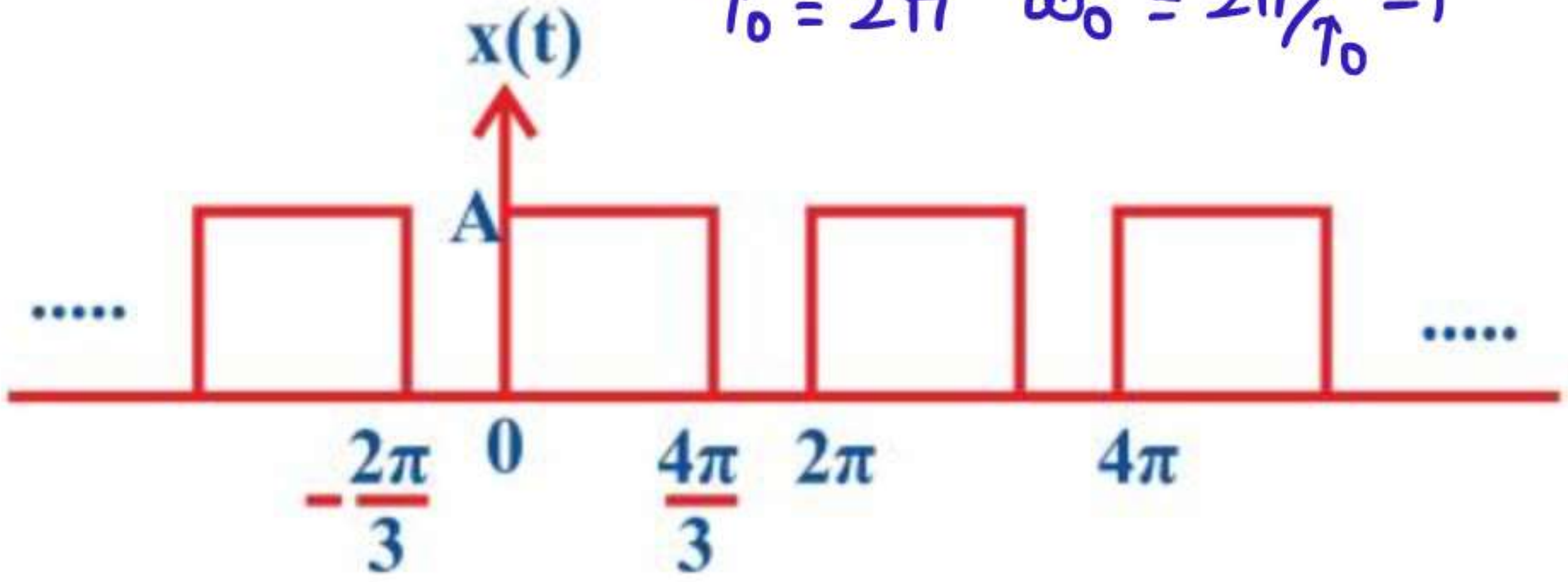
(A) $\frac{A}{2\pi n}\left(e^{-\left(\frac{4\pi n}{3}\right)}-1\right)$

(B) $j\frac{A}{2\pi n}\left(e^{-j\left(\frac{4\pi n}{3}\right)}-1\right)$

(C) $-j\frac{A}{2\pi n}\left(e^{-j\left(\frac{4\pi n}{3}\right)}-1\right)$

(D) $\frac{-A}{2\pi n}\left(e^{-j\left(\frac{4\pi}{3}\right)}-1\right)$

$T_0 = 2\pi \quad \omega_0 = 2\pi/T_0 = 1$



$$C_n = \frac{1}{2\pi}\int_0^{4\pi/3} A\cdot e^{-jn\omega_0 t}\,dt$$

$$= \frac{A}{2\pi}\left[\frac{e^{-jnt}}{-jn}\right]_0^{4\pi/3} = \frac{A}{2jn\pi}\left[1-e^{-j4\pi n/3}\right]$$

$$= -jA/2\pi n\left(1-e^{-j4\pi n/3}\right)$$

Question-11

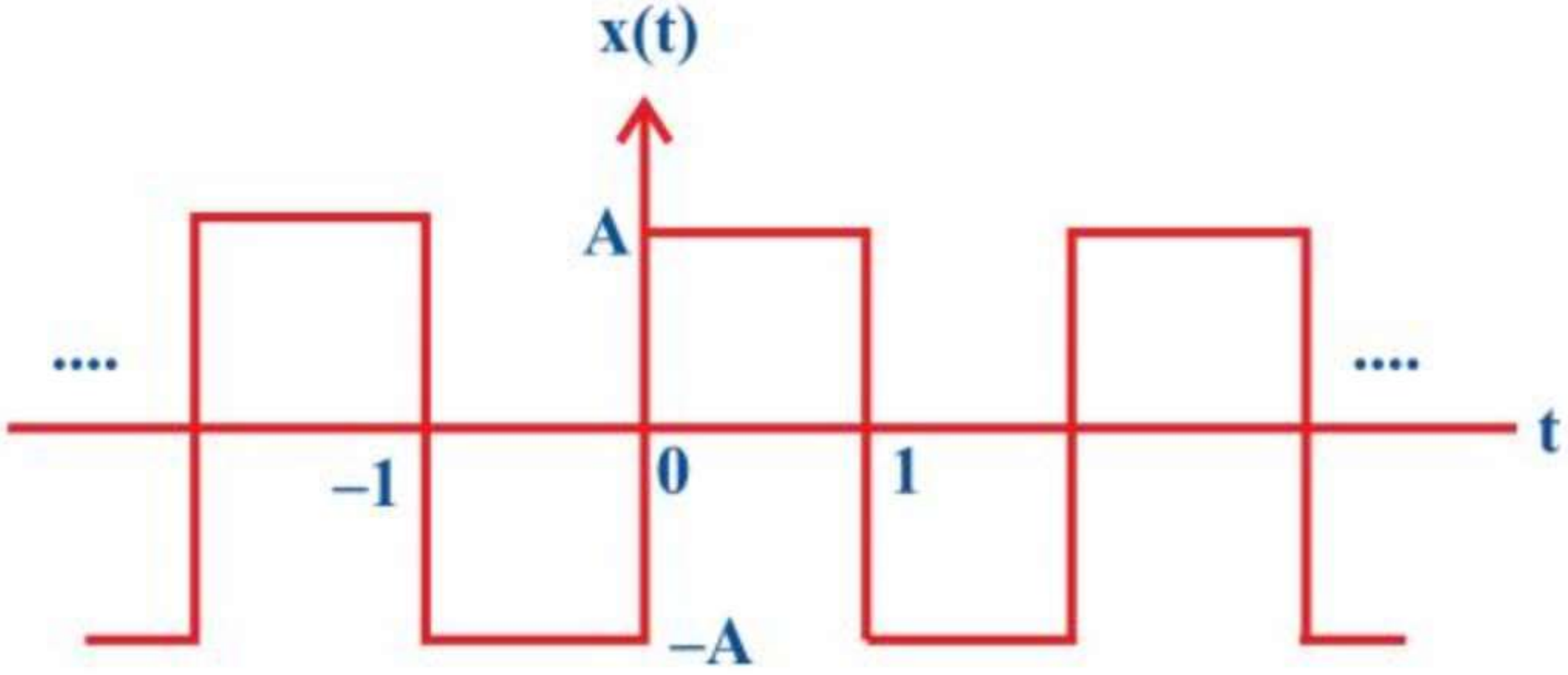
The exponential Fourier series coefficient for the periodic signal shown below is

(A) $\frac{A}{n\pi}[1-(-1)^n]$

(B) $\frac{A}{n\pi}[1+(-1)^n]$

(C) $\frac{A}{jn\pi}[1-(-1)^n]$

(D) $\frac{A}{jn\pi}[1+(-1)^n]$



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